Report suggests high risk in low tar cigarettes

By Richard A. Knox
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Smokers who switch to new low tar and nicotine cigarettes, thinking them safer than their old brands, may be deceiving themselves and possibly even raising their risk of smoking-related disease.

A Harvard study has found that smokers unconsciously alter their smoking behavior when given high tar and low tar cigarettes without knowing which was which. Most smokers in the study consistently held the smoke from the low tar cigarettes in their lungs a longer time in an apparent [illegible]rt to extract more satisfaction from them.

The results, reported last week at a meeting in San Francisco of the American Federation for Clinical Research, constitute some of the first evidence undermining the tobacco industry's recent advertising that implies low tar and ultra low tar cigarettes may be safer.

The study, which involved about 30 smokers over the past two years, was conducted by the Harvard University Tobacco and Health Research Program, a six-year-old, $1 million-a-year project funded by the National Institutes of Health and a group of seven tobacco companies. The project is directed by Dr. Gary L. Huber of the Beth Israel Hospital.

Dr. Ami Rubin, an Israeli researcher working temporarily in Huber's laboratory, reported that regular pack-a-day smokers consistently altered their inhalation patterns when given low tar cigarettes. Their breathing patterns were monitored by an unobtrusive technique without the subjects knowing what was being measured.

Rubin, reporting on part of the results, said that eight out [illegible] 10 smokers held their breath longer when they smoked low tar cigarettes, while two decreased the length of time they held the smoke in their lungs.

"We feel these findings support the hypothesis that nicotine content of the tobacco may modify smoking behavior, and that in our limited study the smokers may titrate [illegible] their nicotine requirement while smoking low nicotine cigarettes by keeping the smoke in their lungs for a longer period of time," Rubin concluded.

The medical director of the Tobacco Institute, a Washington-based [illegible] group for the industry, said that the [illegible] research group [illegible] to confirm other studies which indicate that people smoke for nicotine, and that they self-regulate their own dose levels." But, he added: "So what?"

Dr. Charles Waite, the Tobacco Institute doctor, said he had not seen the research report but commented that it may have been flawed by the small number of smokers involved and the artificial situation in which their smoking behavior was tested.

Waite also said that cigarette companies have not implied in their advertising that low tar cigarettes are safer. "People who worship at the temple of health and those who think that by intervening in their own lifestyle they're going to live longer may believe that," Waite said. "But I don't think there's been any assurance from the companies that the lower tar and nicotine cigarettes are any safer.

"I'm told that the idea of selling something is give people what they want — or what they think they want," Waite added. "The companies are fairly perceptive and if there is a consumer demand, why shouldn't they satisfy it?"

The Harvard project was able to classify smokers into four categories according to their inherent smoking patterns. "Deep inhalers" consistently inhaled puffs that were two to four times as large as their normal breathing volumes at rest. "Breath holders" customarily held the smoke in their lungs for a longer period, similar to the way many people smoke marijuana. "Purgers" invariably empty their lungs almost entirely in a vigorous exhalation. And "shallow inhalers" barely take any smoke into their lungs at all — even if they have been pack-a-day smokers for years.

The existence of this inhalation-exhalation "fingerprint" for each smoker enabled the researchers to spot shifts in smoking behavior tied to the type of cigarette he or she was given.

Rather than use brand name cigarettes, the researchers used two reference cigarettes, one with a tar and nicotine content as high as anything on the market, while the other corresponded to the newest ultra low tar and nicotine brands.

The Harvard researchers are currently pursuing their hunch that the way a smoker inhales is related to his risk of developing lung disease.

Coupled with recent British findings that people who switch to low tar and nicotine brands also tend to smoke more cigarettes to satisfy their nicotine craving, the new finding suggests that "safer" cigarettes may not pose a lower health risk than their high tar and nicotine predecessors.

The effort by US and British cigarette companies to reduce tar and nicotine content in their products goes back nearly a quarter-century. Between 1955 and 1975, the tar content of US cigarette brands was reduced from 43 milligrams per cigarette to 18 milligrams, on the average. More recently, manufacturers have marketed brands containing only 7.3 milligrams of tar and .53 milligrams of nicotine, on average.

Tar and nicotine, as constituents of the "particulate," or solid phase, of tobacco smoke, are linked, that is, when tar is removed, nicotine content goes down too. Most smokers — but not necessarily all — presumably crave the effects of nicotine for its effects on the nervous system.

Newer low tar brands may comprise nearly a quarter of the $20 billion-a-year US cigarette market, with new versions coming out at a fast pace. Following the assumption that they may be less hazardous, Sen. Edward M. Kennedy ([illegible]) and others have proposed that high tar and nicotine cigarettes be taxed at a higher rate to encourage smokers to switch — a proposal that Huber believes is premature. "We don't have enough data to make that assumption," the Harvard researcher said.

The effort to reduce tar and nicotine is predicated on research that demonstrated a higher incidence of cancer in mice after cigarette tar, the sticky residue of tobacco smoke, was painted on the animals' bodies.

There are no human studies suggesting that tar and nicotine alone are the hazardous constituents of cigarette smoke. Some researchers believe that part of the risk may be related to the gas phase of tobacco smoke.

The gas phase contains carbon monoxide, a class of compounds that recently has been shown to disable the cells that attack disease organisms invading the lungs, and possibly also unidentified chemicals that might cause cancer. A group of Swiss scientists demonstrated in 1974 that substances in the gas phase of cigarette smoke caused cancerous changes in human lung tissue grown in laboratory dishes.

The gas phase of even the lowest tar and nicotine brands is virtually the same as that of the older, hypothetically more hazardous brands.

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